

Amirreza “Farnam” Taheri

MSc in Economics · Structural Modeling · Econometrics
Machine Learning & Deep Learning

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Graduate student in Economics at **Tehran Institute for Advanced Studies (TelAS)**. Research in labor economics, empirical econometrics, and structural search-and-matching models supervised by **Prof. Alireza Sepahsalari**. Focused on dynamic structural modeling, microeconometrics, machine learning, and deep learning in Python, Stata, and R, alongside software systems engineering in Rust, Go, and TypeScript.

MASTER’S THESIS RESEARCH

Liquidity, Mismatch, and Job Choice: A Structural Search Model

Thesis

Department of Economics and Finance, TelAS · Advisor: **Prof. Alireza Sepahsalari**

- Continuous-time Bellman equations with endogenous asset accumulation and reservation wage policies.
- Numerical solution via Value Function Iteration (VFI) and continuous-state dynamic programming.
- Data science & econometric analysis:** Microdata analysis and moment estimation with NLSY79 and O*NET; reported official survey variable title errata (W1382000, W1391400), confirmed by OSU CHRR archivist.

Microdata: NLSY79, O*NET, CPS · Stack: Python, Stata, R, Dynamic Programming, LaTeX

GRADUATE & MINOR COURSEWORK

Econometrics I

19.3 / 20

Tools: Stata, LaTeX · [GitHub](#)

Classical Linear Regression · Asymptotic Theory · Hypothesis Testing · Instrumental Variables (IV/2SLS)

Econometrics II (Applied Econometrics)

19.0 / 20

Tools: R, R Markdown · [GitHub](#)

Generalized Method of Moments (GMM) · Panel Data & Fixed Effects · DiD & Synthetic DiD · RDD · IV

Macroeconomics I

19.0 / 20

Tools: Python, MATLAB, Dynare, LaTeX · [GitHub](#)

Stochastic RBC Simulation · Money & Govt in RBC · Job Search (McCall & DMP) · Dynamic Programming

Machine Learning

18.5 / 20

Tools: Python, scikit-learn, NumPy, Pandas · [GitHub](#)

Supervised Learning & Regularization · SVM & Kernel Methods · Tree Ensembles · Time-Series · Clustering

Neural Networks & Deep Learning

17.0 / 20

Tools: Python, PyTorch, Transformers, LoRA · [GitHub](#)

Backprop & Optimization · CNNs · Sequential Models (RNN, GRU, BiLSTM) · Transformers · RL (DQN) · LoRA

Selected Graduate Courses:

- Microeconomics II (Game Theory): **20.0 / 20** · Mathematics for Economists: **19.3 / 20**
- Macroeconomics II: **19.0 / 20** (Python · MATLAB) · Contract Theory: **18.5 / 20**
- Microeconomics I: **18.1 / 20**

Selected Undergrad Courses:

- Econometrics (Minor): **20.0 / 20** (Stata) · Macroeconomics (Minor): **20.0 / 20**
- Game Theory (Minor): **20.0 / 20** · Research Method & Report Writing: **19.5 / 20**
- Calculus II: **17.6 / 20** · Engineering Economics: **17.0 / 20** · Numerical Analysis: **16.5 / 20**

FEATURED PROJECTS

Scriptor · Local-first Research & Knowledge Workspace

[Repo](#)

Stack: Rust 1.96, React 19, TypeScript, Tauri 2, SQLite FTS5, Pandoc, MCP, Git

- Tauri 2 architecture with Rust kernel core and React 19 / TypeScript renderer.
- SQLite WAL + FTS5 indexing, HMAC-authenticated local RPC, and ts-rs contracts.
- Deterministic Pandoc export (HTML, PDF, DOCX, LaTeX, ePub) and native 3-way Git merge.
- Automated release engineering with CycloneDX SBOM, SHA-256 provenance, Playwright E2E.

WinCare · System Diagnostics & Telemetry Platform

[Repo](#)

Stack: WinUI 3, .NET 8, C#, Rust 2024, ONNX Runtime, DirectML

- Hybrid .NET 8 / WinUI 3 architecture with high-performance Rust core engine.
- Local ONNX Runtime inference with DirectML hardware acceleration for anomaly detection.
- Windows Job Objects process isolation, telemetry inspection, and policy-governed maintenance.

CORE TECHNICAL STACK

Structural Modeling & Econometrics

Continuous-Time Bellman Equations, Dynamic Programming, Value Function Iteration (VFI), SMM / Structural Estimation, Panel Data Econometrics, GMM, 2SLS / Instrumental Variables, Nonparametric Smoothing, Dynare / MATLAB, Stata, Python, R, LaTeX

Data Science & Data Engineering

Survey Microdata Processing (NLSY79, O*NET, HEIS, LFS), SQL, ETL Data Pipelines, Pandas, NumPy, SciPy, Data Cleaning & Imputation, Nonparametric Smoothing

Machine Learning & Deep Learning

PyTorch, TensorFlow, scikit-learn, Transformers, LoRA Instruction Tuning, Reinforcement Learning (DQN), Time-Series Forecasting, AI Agents / Agentic Coding

Systems & Software Engineering

Rust 2024, Go, TypeScript, React 19, Tauri 2, WinUI 3 / C# .NET 9, SQLite FTS5, Docker, Git, GitHub Actions, Linux, CI/CD

EDUCATION & HONORS

M.Sc. in Economics

18.72 / 20

Tehran Institute for Advanced Studies (TelAS)

Sep 2023 — Present

Thesis advisor: **Prof. Alireza Sepahsalari**

Economics Minor

19.20 / 20

Amirkabir University of Technology (AUT)

Sep 2018 — Sep 2023

Completed alongside engineering degree. Key courses: Econometrics (20.0), Macro (20.0), Game Theory (20.0).

B.Sc. in Civil Engineering

15.69 / 20

Amirkabir University of Technology (AUT)

Sep 2018 — Sep 2023

HONORS & DISTINCTIONS

★ Rank 27 — Master’s Konkur

2023

Nationwide entrance exam for Masters in Economics.

★ TelAS Merit Scholarship

2023 & 2025

Competitive full graduate merit scholarship awarded by TelAS.